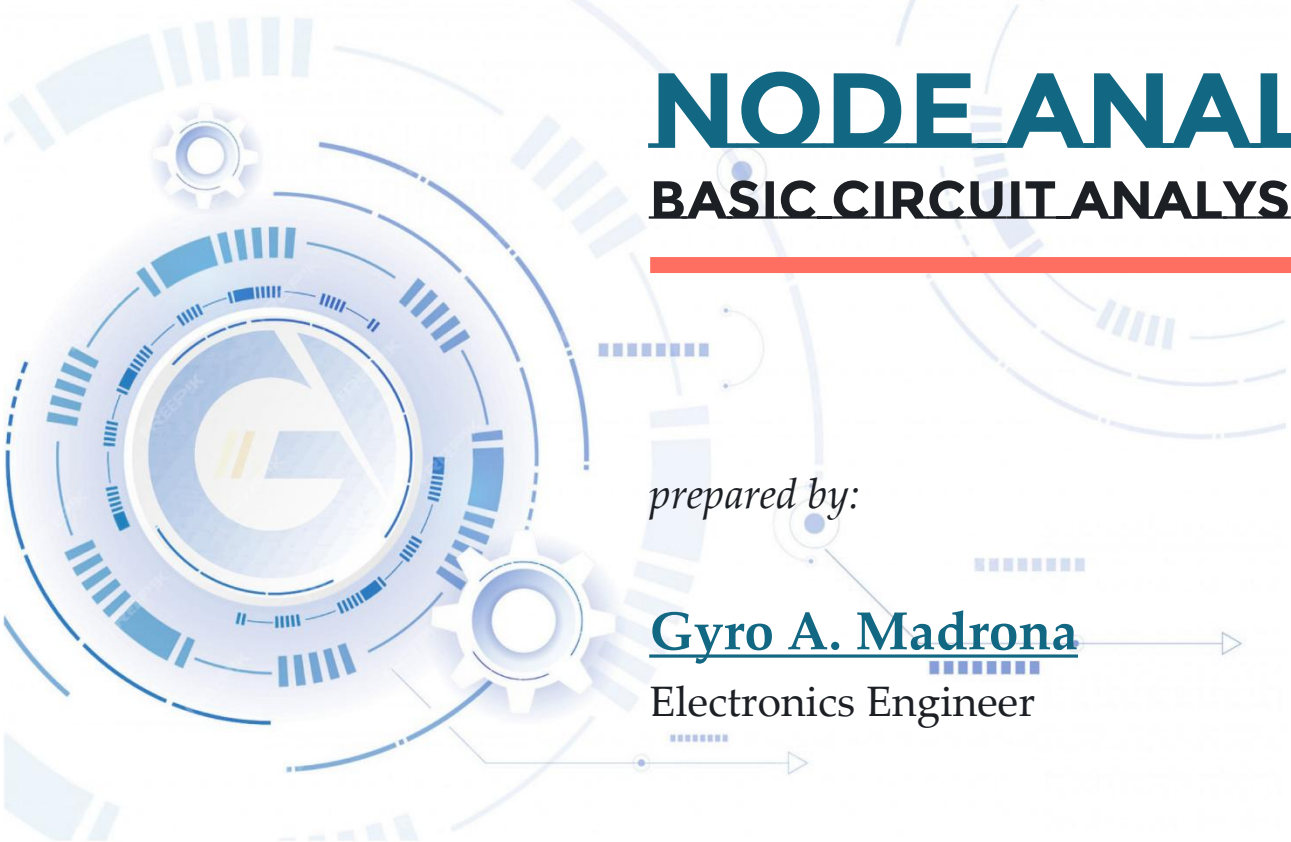




NODE ANALYSIS METHOD

BASIC CIRCUIT ANALYSIS METHOD

prepared by:

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TOPIC OUTLINE

Node Voltage

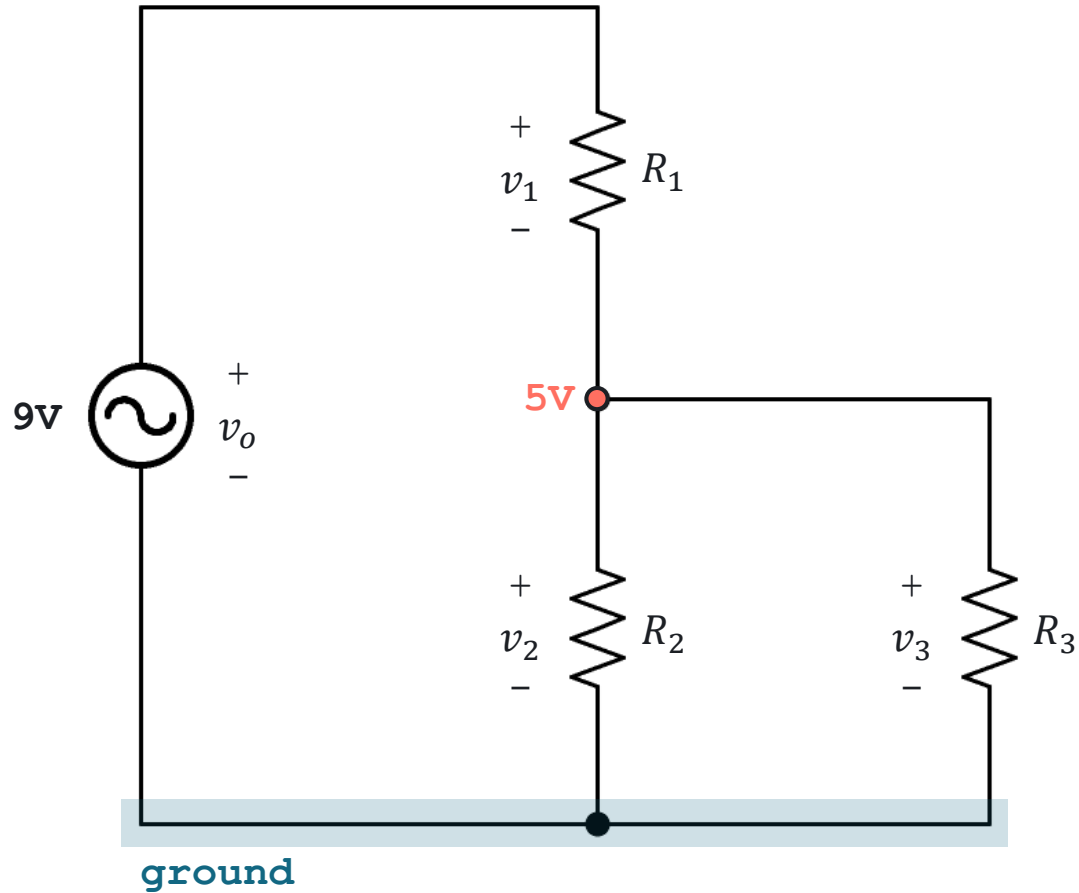
Node Analysis Method



NODE ANALYSIS METHOD



NODE VOLTAGE



Node Voltage is the electrical potential difference at a specific node in a circuit relative to a reference node.

Example

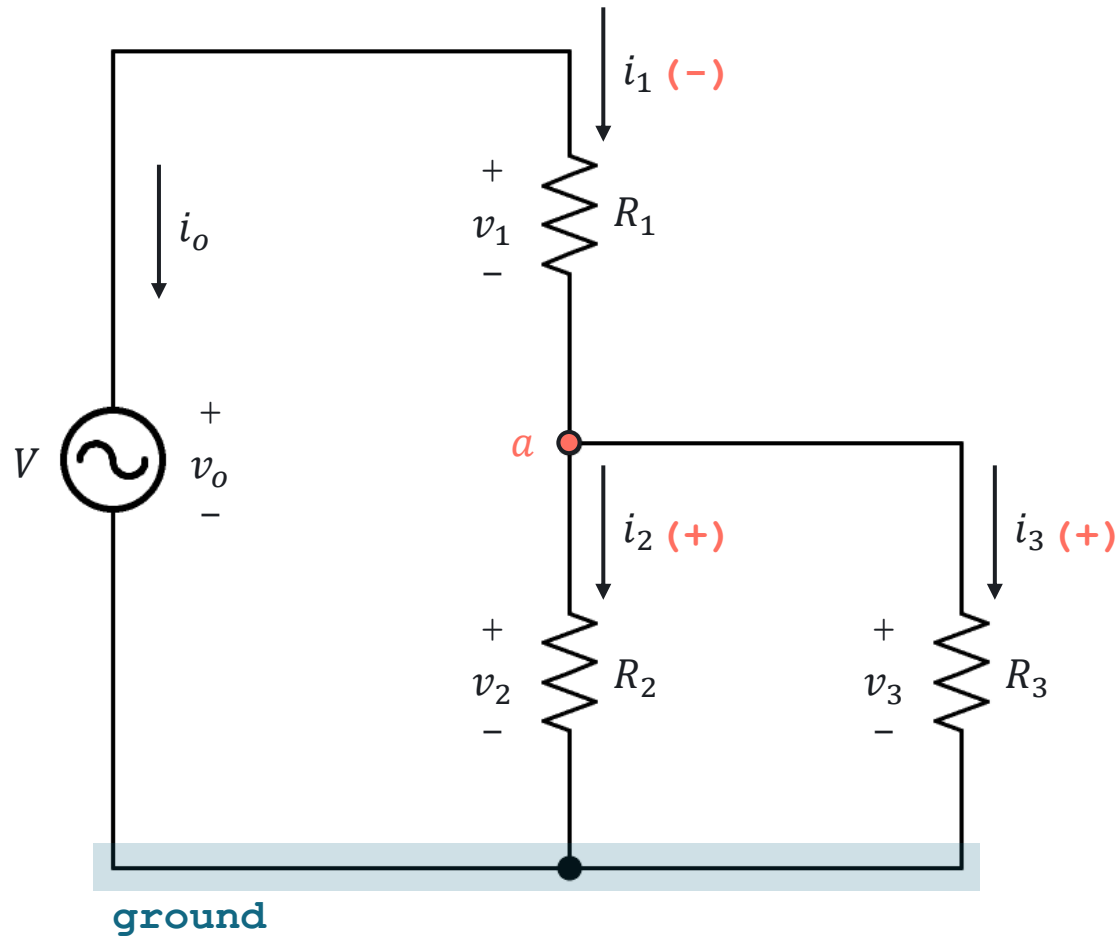
$$v_1 = 9V - 5V$$

$$v_2 = 5V - 0$$

$$v_3 = 5V - 0$$



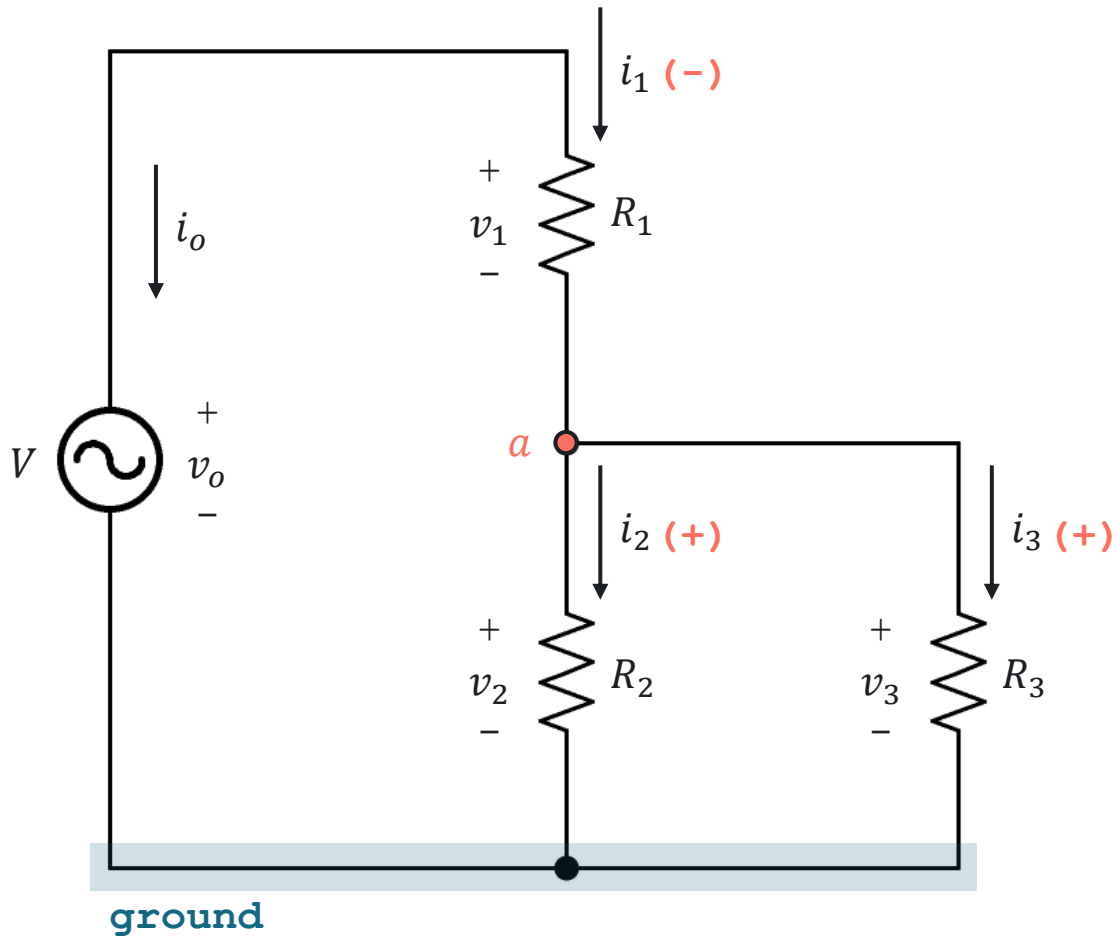
NODE ANALYSIS METHOD



The node analysis method is based on Kirchhoff's current law (KCL), which is implicitly applied to establish voltage-current relationships forming a system of equations to be solved for the unknown node voltages.



NODE ANALYSIS METHOD



KCL @a

$$-i_1 + i_2 + i_3 = 0$$

$$-\frac{v_1}{R_1} + \frac{v_2}{R_2} + \frac{v_3}{R_3} = 0$$

$$-\frac{v_o - v_a}{R_1} + \frac{v_a - 0}{R_2} + \frac{v_a - 0}{R_3} = 0$$

$$-v_o G_1 + v_a G_1 + v_a G_2 + v_a G_3 = 0$$

$$v_a(G_1 + G_2 + G_3) = v_o G_1$$

$$v_a = \frac{v_o G_1}{G_1 + G_2 + G_3}$$



ELECTRICAL POWER

Electrical power refers to the rate at which electrical energy is converted per unit time (joules/second).

Formulas

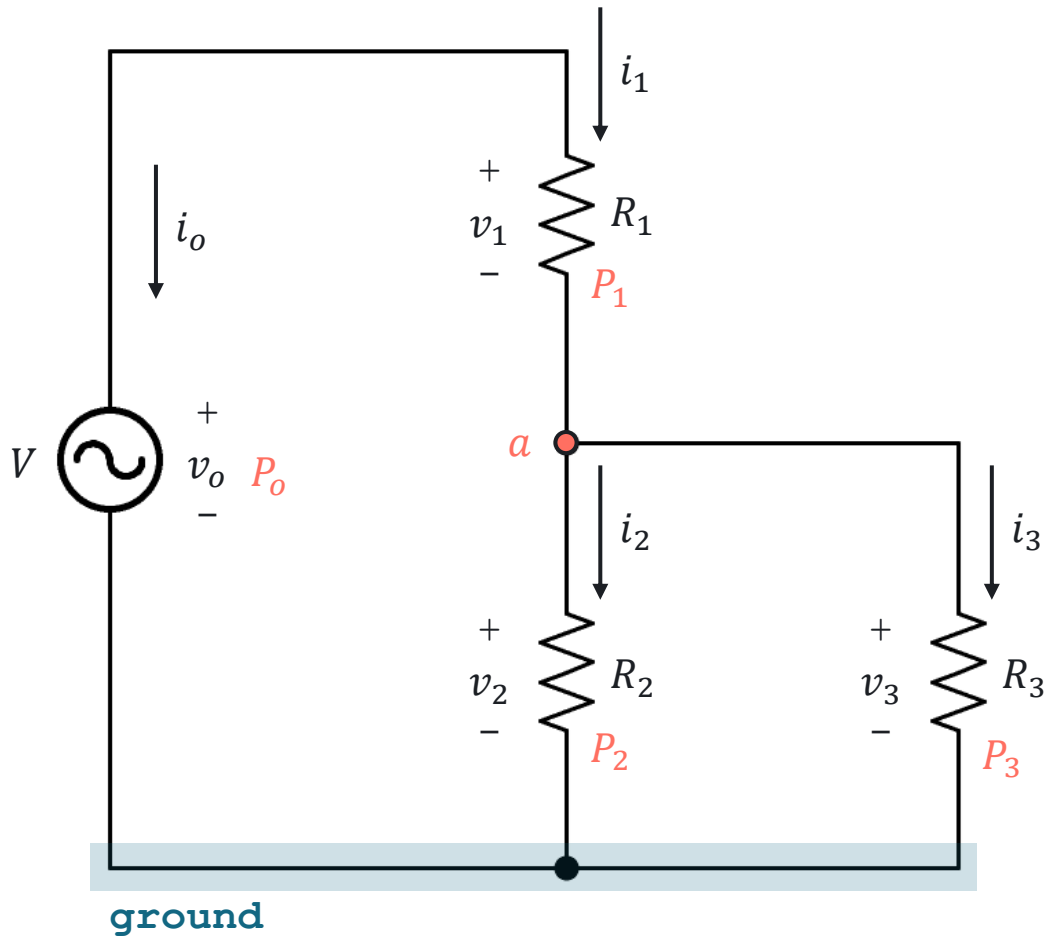
$$P = vi$$

$$P = i^2 R$$

$$P = \frac{v^2}{R}$$

$$P_o = P_1 + P_2 + P_3 + \cdots P_n$$

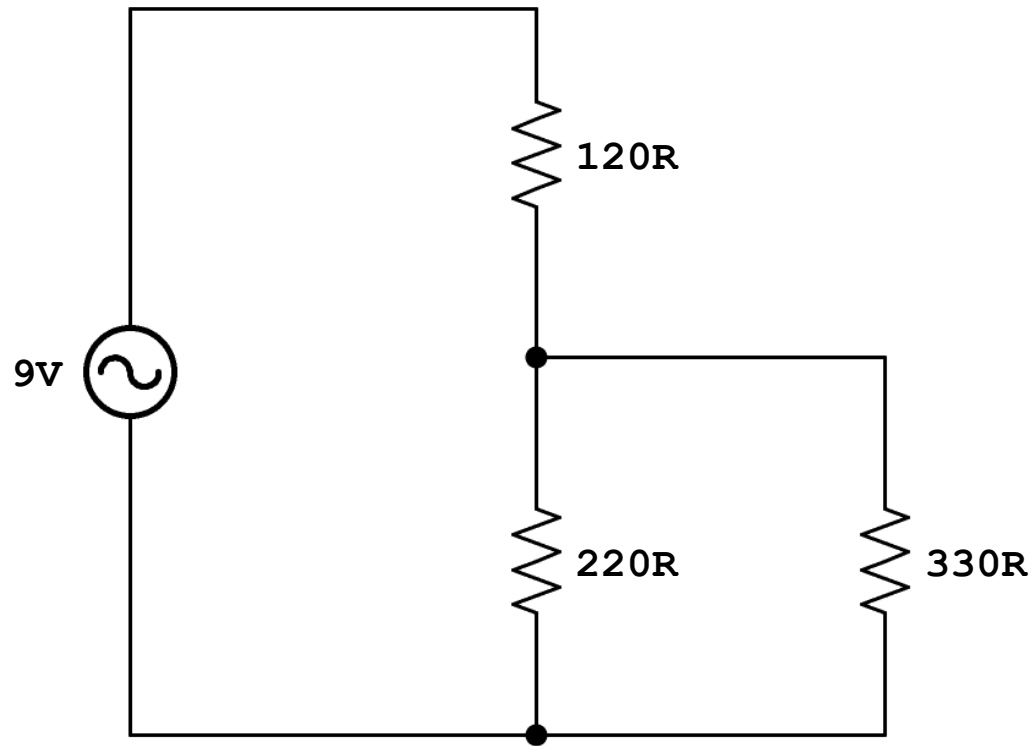
unit: Watt (W)



EXERCISE

Determine the current through and voltage across each resistor, as well as the total power of the given circuit.

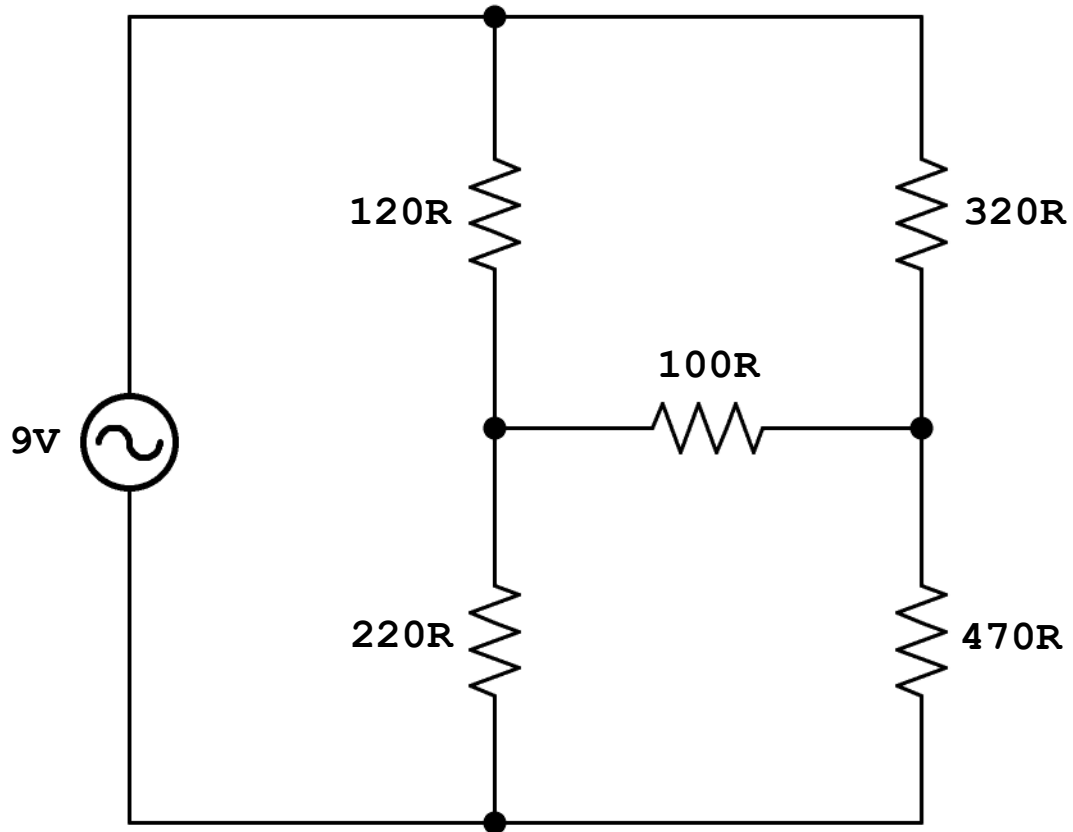
Solution



EXERCISE

Determine the current through and voltage across each resistor, as well as the total power of the given circuit.

Solution



LABORATORY

